

Qwen3-4B serving benchmark: SGLang + FlashInfer

Qwen3-4B · NVIDIA A100-SXM4-40GB, 580.178.04, 40960 MiB · SGLang 0.5.21 · FlashInfer 0.6.18 · torch 2.13.0+cu130 (CUDA 13.0) · started 2026-10-05T21:14:36 · SLO at p99: TTFT ≤ 500 ms, ITL ≤ 50 ms

MAX AGENT CONCURRENCY WITHIN SLO

1

108 output tok/s · TTFT p99 221 ms

PEAK OUTPUT THROUGHPUT

1,576 tok/s

agent workload at c=64

MAX SUSTAINED QPS WITHIN SLO

none

lowest offered rate already missed the SLO

TTFT P50 / P99 / P99.9

989 / 4,030 / 4,031 ms

400 requests, agent workload, c=64

ITL P50 / P99 / P99.9

31.0 / 39.1 / 549.7 ms

101,994 inter-token gaps

BURST: FAILED REQUESTS

0 of 304

all sent at once · TTFT p99 6,015 ms

ERRORS ACROSS ALL TESTS

0 of 2,082

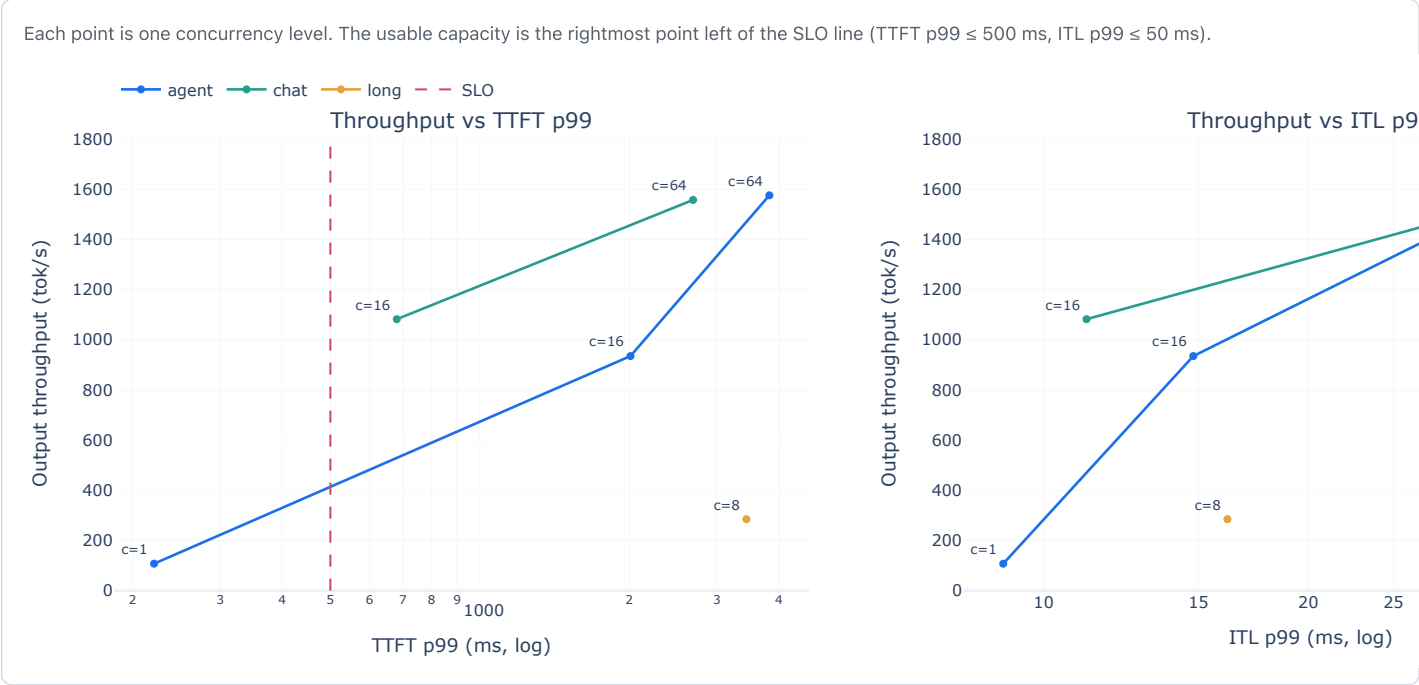
0.000% of requests

Settings vs baseline

At the highest concurrency every setting ran; change vs baseline in brackets. Lower is better for latency.

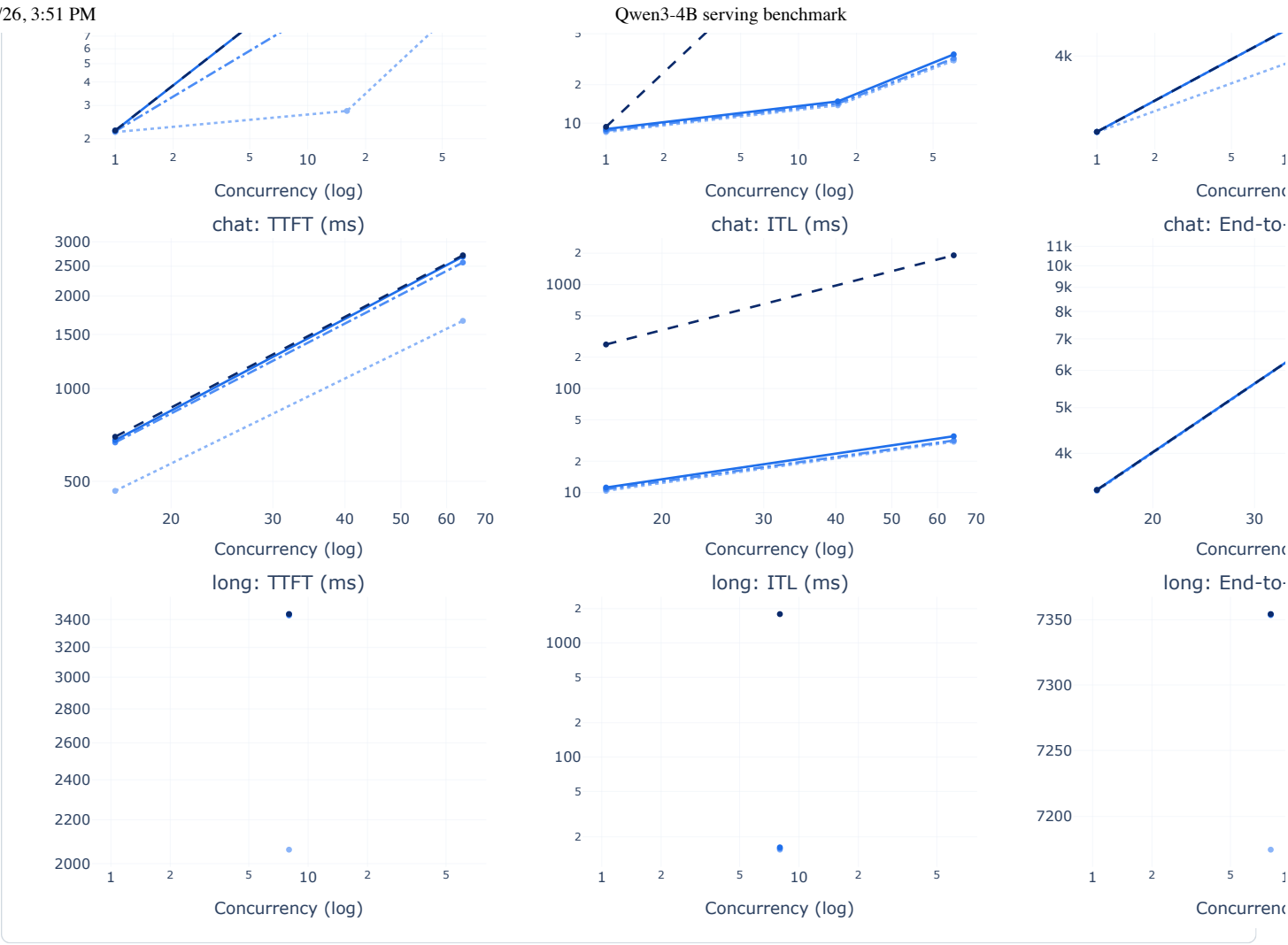
setting	workload	concurrency	output tok/s	TTFT p99 ms	ITL p99 ms
no_radix	agent	64	641 (-59%)	20,059 (+424%)	40.6 (+18%)
fp8_kv	agent	64	1,491 (-5%)	3,920 (+2%)	40.5 (+17%)
page16	agent	64	1,567 (-1%)	3,851 (+1%)	34.0 (-1%)
triton	agent	64	1,313 (-17%)	4,620 (+21%)	43.8 (+27%)

Capacity: throughput vs latency (baseline)

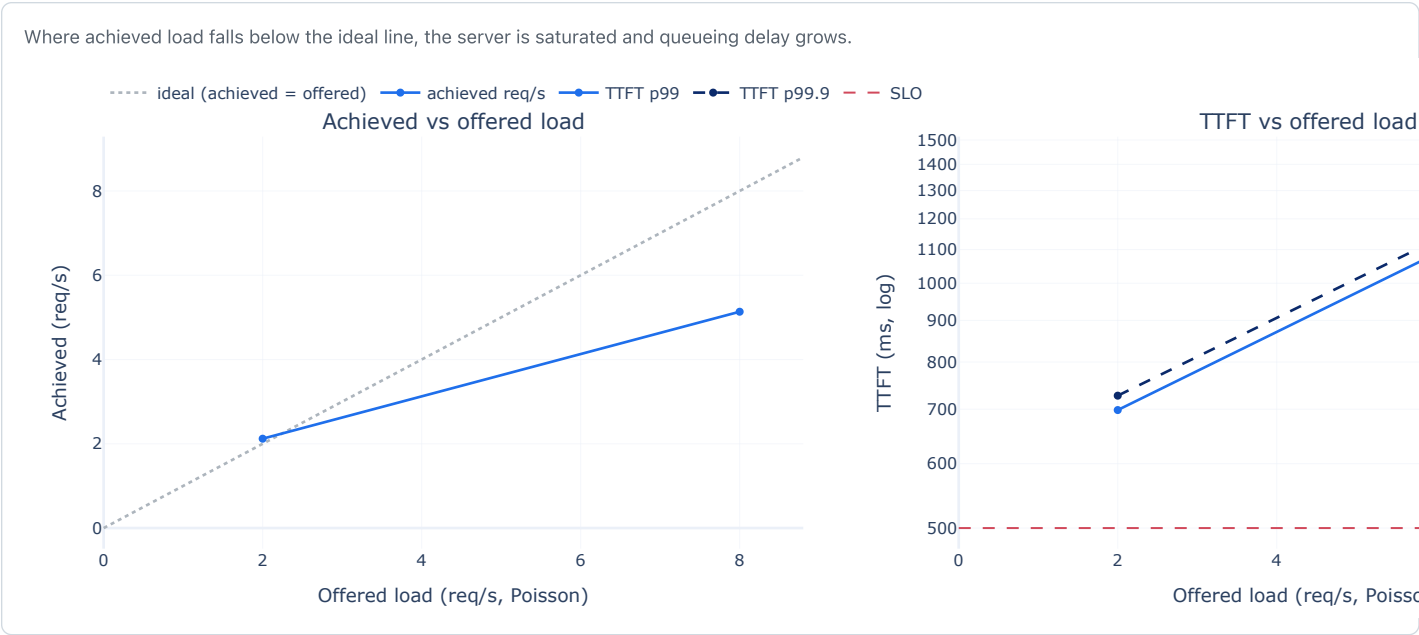


Latency percentiles vs concurrency (baseline)



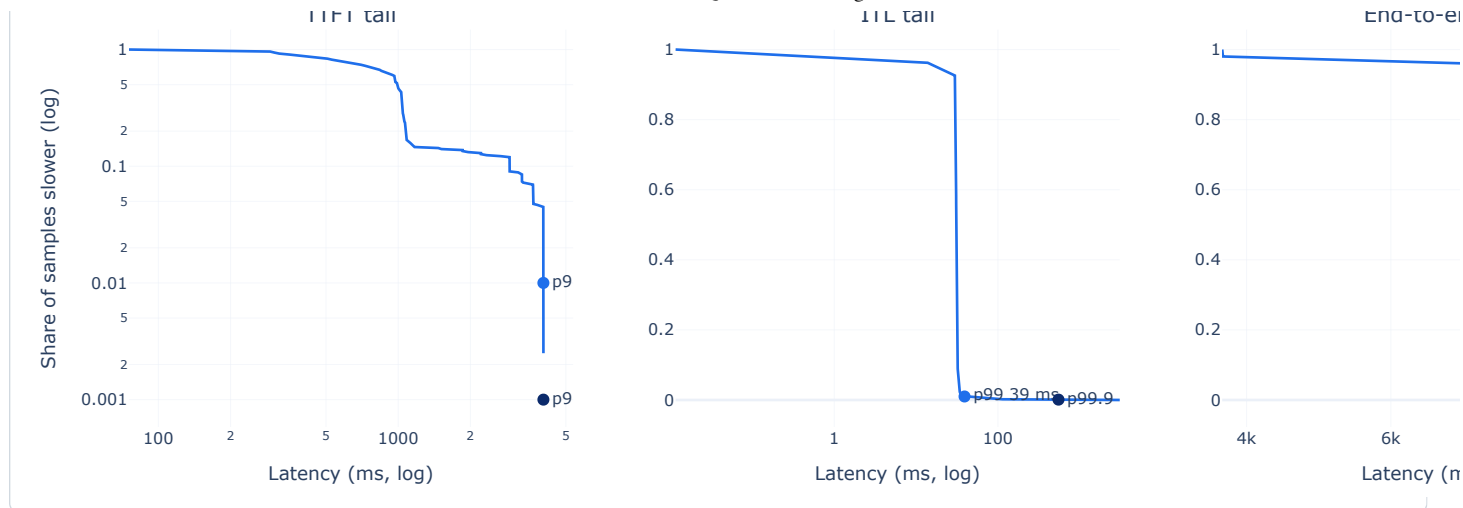


QPS sweep: Poisson arrivals (baseline, agent workload)



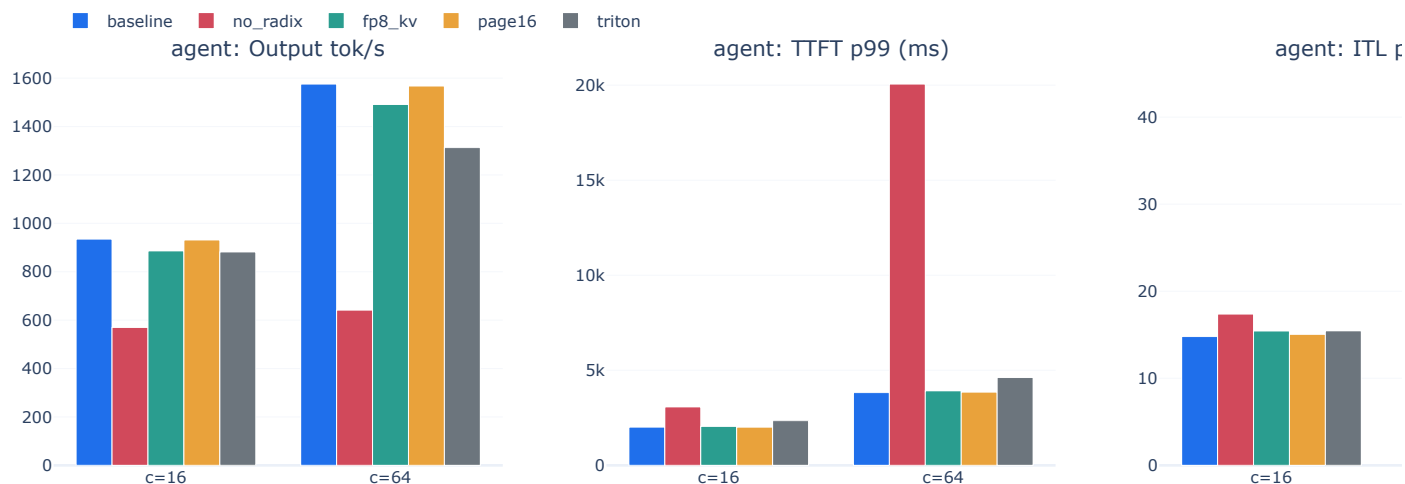
Tail latency at the operating point (p99.9 run)

Survival curve: the share of samples slower than each latency. p99.9 TTFT rests on the number of requests in this run; ITL percentiles use every inter-token gap.



KV-cache and FlashInfer settings: head-to-head

Same workload and concurrency for every setting; only the server flag changes.

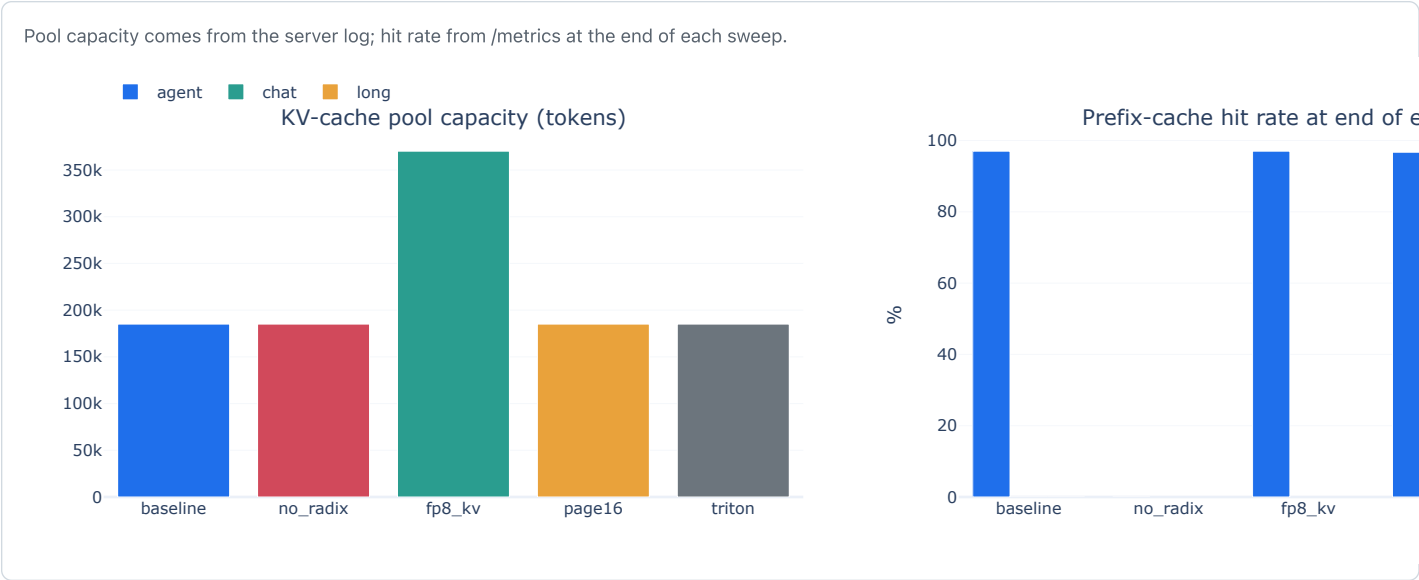


KV-cache and FlashInfer settings: capacity curves

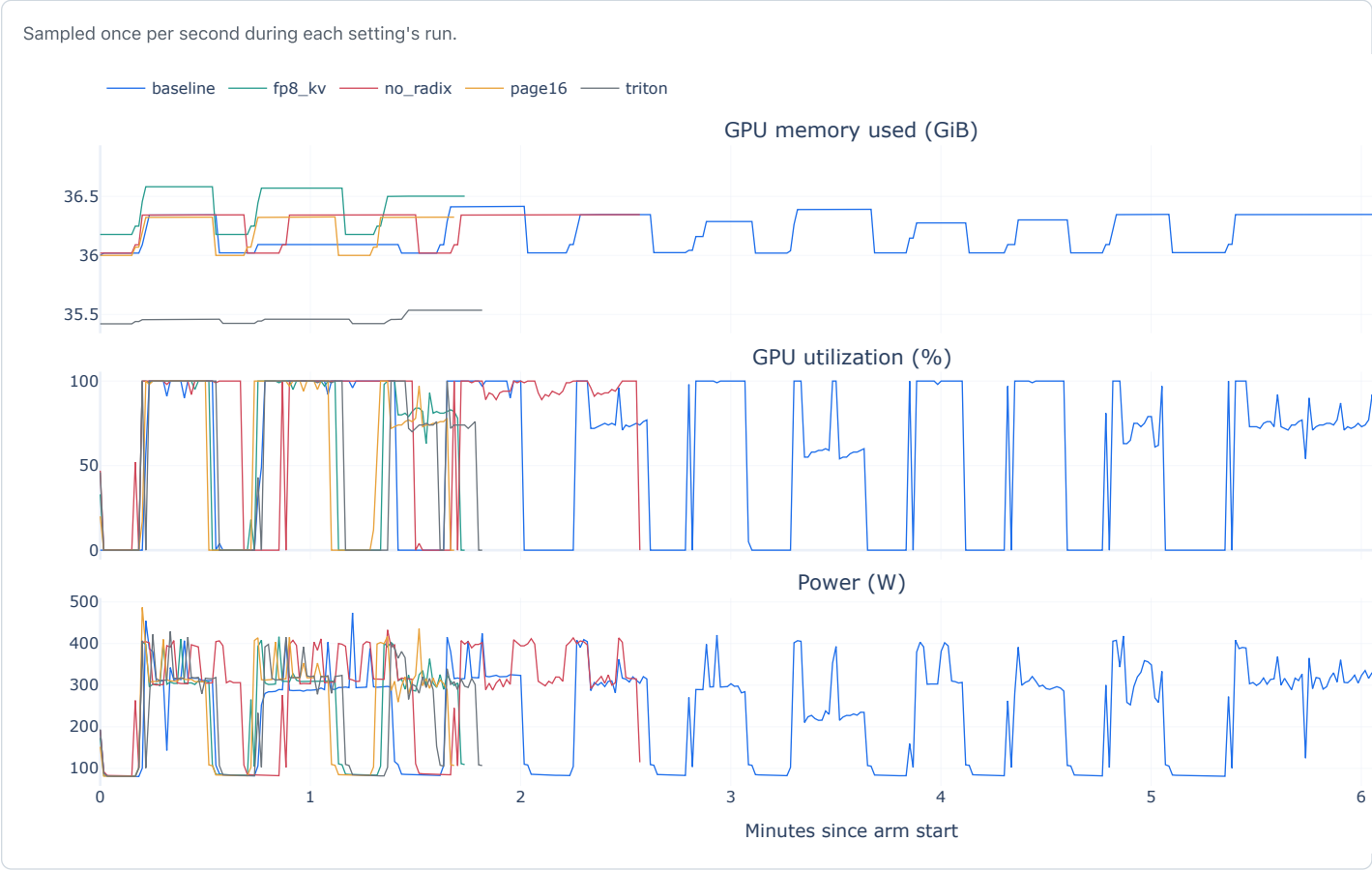
A curve further up and to the left serves more tokens at lower latency.



KV-cache capacity and prefix-cache hit rate



GPU telemetry



All test points

Latencies in ms, computed from per-request data. n_tftf is the number of requests behind each TTFT percentile; treat p99.9 with fewer than 1,000 requests as indicative.

arm	workload	mode	value	num_prompts	failed	req_s	out_tok_s	ttft_p50	ttft_p90	ttft_p99	ttft_p999	itl_p50	itl_p90	itl_p99	itl_p999	e2e_p50
baseline	agent	burst	300	304	0	12.0	3062.7	4997.2	5808.4	6015.1	6096.4	75.0	80.2	96.6	1251.1	25160.0
baseline	agent	conc	1	16	0	0.4	107.5	216.8	218.9	220.8	221.1	8.5	8.7	9.0	9.4	2389.0
baseline	agent	conc	16	80	0	3.7	934.8	280.7	1192.6	2009.9	2010.0	13.8	14.2	14.8	476.1	4222.0
baseline	agent	conc	64	128	0	6.2	1575.5	1051.9	3820.0	3827.3	3854.9	30.9	32.1	34.5	590.3	10372.0

arm	workload	mode	value	num_prompts	failed	req_s	out_tok_s	ttft_p50	ttft_p90	ttft_p99	ttft_p999	itl_p50	itl_p90	itl_p99	itl_p999	e2e_p50
baseline	agent	p999	64	400	0	6.5	1672.2	988.8	2914.3	4029.6	4030.7	31.0	32.2	39.1	549.7	9102.7
baseline	agent	rate	2	32	0	2.1	543.0	226.4	519.5	698.3	727.4	13.3	16.4	197.1	217.1	4536.3
baseline	agent	rate	8	80	0	5.1	1314.3	99.1	1138.8	1355.8	1408.4	31.2	45.7	94.2	373.7	8939.8
baseline	chat	conc	16	66	0	4.2	1081.8	465.4	668.7	680.3	697.8	10.4	10.8	11.2	265.2	3330.4
baseline	chat	conc	64	128	0	6.1	1557.0	1661.2	2570.8	2686.9	2712.4	30.8	31.6	34.8	1906.7	10517.8
baseline	long	conc	8	16	0	1.1	284.8	2061.7	3430.7	3440.1	3440.3	15.4	15.8	16.2	1790.4	7175.0
no_radix	agent	conc	16	80	0	2.2	569.8	1783.5	3007.3	3074.9	3088.8	16.1	16.5	17.4	1915.3	7170.8
no_radix	agent	conc	64	128	0	2.5	641.3	7430.3	18974.3	20059.0	22326.1	29.9	31.1	40.6	5097.0	15206.0
fp8_kv	agent	conc	16	80	0	3.5	886.2	299.7	1212.4	2047.5	2047.7	14.6	15.0	15.4	518.4	4476.8
fp8_kv	agent	conc	64	128	0	5.8	1491.2	1107.6	3910.7	3920.4	3949.8	32.7	34.0	40.5	644.6	10961.4
page16	agent	conc	16	80	0	3.6	931.3	288.6	1196.0	2007.9	2008.1	13.8	14.2	15.0	474.7	4254.8
page16	agent	conc	64	128	0	6.1	1567.5	1058.0	3848.6	3851.4	3884.6	31.0	31.9	34.0	676.3	10398.4
triton	agent	conc	16	80	0	3.4	881.8	357.1	1376.8	2351.4	2351.6	14.1	14.5	15.5	628.4	4468.8
triton	agent	conc	64	128	0	5.1	1313.0	2933.0	4612.2	4619.9	4650.3	33.7	35.3	43.8	826.3	12452.6

Method

- **baseline:** FlashInfer, radix cache on, BF16 KV, page size 1 (baseline)
- **no_radix:** Prefix (radix) cache off `--disable-radix-cache`
- **fp8_kv:** FP8 e5m2 KV cache `--kv-cache-dtype fp8_e5m2`
- **page16:** KV page size 16 `--page-size 16`
- **triton:** Triton attention, no FlashInfer `--attention-backend triton --sampling-backend pytorch`

Workloads: agent = 16 shared 4K-token system prompts + 128-token question, 256 output tokens; chat = 1K in / 256 out; long = 8K in / 256 out. Fixed output length, prefix cache flushed before every point, load from `sglang.bench_serving` on the same host.

▶ baseline launch command

▶ fp8_kv launch command

▶ no_radix launch command

▶ page16 launch command

▶ triton launch command